

Quantum Computing

Part 4 - Hardware by Modality: Photonic Quantum Computing

Compiled by the Spaced Research Ledger

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Photonic quantum computers use particles of light as their qubits. Light does not need to be cooled, and does not pick up noise from its surroundings the way matter does. It travels through ordinary optical fibre, so a photonic machine can be spread across several racks or even several rooms. Most of the systems in this part run at room temperature apart from their detectors.

The problem is that photons get lost. A photon that scatters or gets absorbed simply is not there any more. Unlike the errors in Parts 1 through 3, loss is not a state that can be corrected back. Every architecture here is organized around that single failure mode, and the fidelity figures below are quoted for photons that arrived.

A second thing to watch is what kind of machine is being described. Some of these systems are universal and gate-based, meaning they can run any quantum algorithm. Others perform one fixed sampling task extremely well and cannot be programmed for general work. Section 5 covers the largest machines in this part, and they are the second kind.

Section 1 covers PsiQuantum, Section 2 Xanadu, Section 3 Quandela, Section 4 ORCA Computing, and Section 5 the University of Science and Technology of China.

1. PsiQuantum

PsiQuantum is building for a semiconductor foundry from the start, and its architecture is designed around tolerating photon loss rather than avoiding it. It has published detailed component fidelities and is unusually explicit about what those numbers exclude.

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The architecture is called fusion-based quantum computing [1]. Rather than using conventional entangling gates, it performs entangling measurements called fusions on qubits drawn from small, constant-sized entangled resource states. The advantage is that resource states stay small no matter how large the computation grows.

The Omega platform integrates superconducting single-photon detectors, single-photon sources, and a high-performance optical switch onto a single ultra-low-loss silicon nitride platform [2]. PsiQuantum manufactures waveguide-integrated, photon-number-resolving detectors at GlobalFoundries, using superconducting films biased near their transition point [2]. It also developed a 300-millimetre process for fast, low-loss optical switches based on barium titanate, a material the company calls the holy grail of optical switching [2]. Switching matters because photon sources and gates are probabilistic, so the machine must reroute around operations that failed.

The component fidelities were published in Nature in February 2025, from a platform fabricated at GlobalFoundries' 300-millimetre foundry [3]. State preparation and measurement fidelity was $99.98\% \pm 0.01\%$. Interference visibility between independent photon sources was $99.50\% \pm 0.25\%$. Two-qubit fusion fidelity was $99.22\% \pm 0.12\%$. Chip-to-chip interconnect fidelity was $99.72\% \pm 0.04\%$ across 42 metres of fibre.

That last figure is the one to notice. A photonic machine can be split across chips connected by ordinary fibre, which is not possible for any other modality in this collection.

One qualification governs all four numbers. They are conditional on successful photon detection and do not account for optical loss [3]. They describe how well the machine handles photons that arrived, not how many arrived.

Loss is handled instead by the error-correction budget. The architecture tolerates up to a 10.4% probability of photon loss per fusion, equivalent to about 2.7% loss per photon [4]. It demonstrates a 43.2% threshold against fusion failure using a ballistic technique. The previously reported figure was 14.9%.

2. Xanadu

Xanadu builds its qubits out of squeezed light rather than individual photons, and assembled its current machine from commercially available parts. Its architecture is the most modular in this part, and the company is candid that optical loss is its main limit.

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Aurora was unveiled in January 2025 and published in Nature [5]. It is a 12-qubit universal, gate-based photonic quantum computer built from 35 integrated photonic chips, spread across four modular server racks and connected by 13 kilometres of optical fibre. It runs at room temperature apart from its cryogenic detector system [6].

The qubit count deserves attention against that hardware. Twelve qubits from 35 chips and 13 kilometres of fibre is a demonstration of architecture, not of computational power. What it shows is that the pieces can be connected at all.

The chips were built with established, commercially available fabrication rather than a bespoke foundry process [5]. That is the opposite of PsiQuantum's approach in Section 1, and the two companies are effectively running a controlled experiment on whether custom fabrication is necessary.

The qubits themselves are unusual. Xanadu uses Gaussian boson sampling as a tool to synthesize a bosonic error-correcting state known as a Gottesman-Kitaev-Preskill state, out of squeezed light [7]. The synthesis is probabilistic, so measurements from multiple samplers select the best states to route forward.

The company identifies optical loss as the main source of errors affecting its computational performance, and is focused on driving it down [7].

Xanadu's earlier Borealis processor, a 216-mode squeezed-light interferometer, demonstrated quantum computational advantage through Gaussian boson sampling in a 2022 Nature paper [8]. It has since been retired and superseded by Aurora.

3. Quandela

Quandela's distinguishing technology is a deterministic single-photon source, which addresses the probabilistic-source problem at its origin rather than compensating for it downstream. Its machines are also the most commercially accessible in this part.

Recent Developments

Quandela demonstrated photonic quantum reservoir processing for machine learning and quantum tomography on its Belenos processor in June 2026 [9]. The work used a 24-mode programmable silicon photonic chip with quantum-dot single-photon sources.

The 12-qubit Belenos system was integrated into OVHcloud's quantum platform in April 2026, giving cloud-based access to photonic quantum computing [10].

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Quandela's computers use deterministic single-photon sources based on semiconductor quantum-dot emitters, for linear-optical gate-based computing [11]. Its Prometheus product is a standalone quantum-dot single-photon source that generates and detects photons at record rates, supplying a stable stream of highly indistinguishable photons [12]. Indistinguishability is what allows photons to interfere correctly, and it is the property a good source exists to deliver.

The systems have grown quickly. Belenos, a 12-qubit machine unveiled in May 2025, succeeded the earlier 6-qubit MosaiQ system, and the company says it delivers roughly 4,000 times more computing power [13]. In October 2025 Quandela delivered its 12-qubit Lucy system to the French atomic energy commission [11]. It was installed under the EuroHPC Joint Undertaking, with capacity to scale to 24 qubits by 2026.

Manufacturing is in house. Quandela opened a dedicated cleanroom for producing its own single-photon sources in spring 2024 [14]. It had opened an assembly and online-service facility in Massy, France, in June 2023, the first of its kind in Europe.

4. ORCA Computing

ORCA builds the most conventional-looking hardware in this part. Its systems use telecom components, run at room temperature in standard racks, and are already installed with defence and research customers.

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The architecture uses time-bin-encoded photons stored in fibre-based quantum memory loops [15]. It implements a programmable boson-sampling approach on telecom-compatible components that need no cryogenic cooling.

The PT-2 system began shipping to customers in early 2025 [16]. It expanded the photonic qubit count to about 90 through greater temporal multiplexing and more modes. It runs at room temperature in a standard

19-inch rack and integrates with NVIDIA's CUDA-Q platform. Existing PT-1 units already in the field can be upgraded to PT-2 specifications by adding modules [16].

Deployment came early for this company. ORCA delivered nine PT-1 systems to external customers between 2022 and 2023 [16]. Recipients included the UK Ministry of Defence, the Poznań supercomputing centre, and the UK National Quantum Computing Centre, which used it as a testbed.

One forward-looking claim needs care. In October 2025 ORCA published research describing a nonlinear photonic architecture for fault-tolerant quantum computing, proposing an on-demand optical nonlinearity that lets photons interact at room temperature [17]. This is the company's own research finding rather than an independently verified result.

5. USTC and Gaussian Boson Sampling

The machines in this section are the largest photonic systems ever built, and they are not general-purpose computers. They perform one sampling task, and they perform it at a scale no classical computer can follow.

Recent Developments

China Telecom Quantum Group launched the Tianyan-P2000 photonic quantum computer in June 2026 [18]. It is based on the Jiuzhang 4.0 architecture, controls 2,682 photons, and is connected to the Tianyan quantum cloud platform. A sampling machine reaching a commercial cloud platform is a new development for this modality.

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Jiuzhang 4.0, led by Jian-Wei Pan and Chao-Yang Lu, injects 1,024 high-efficiency squeezed states into a programmable photonic circuit of 8,176 modes [19]. It produces samples with detection events of up to 3,050 photons, an order of magnitude more than previous demonstrations. Source efficiency reached 92%, with 51% overall system efficiency [19].

The classical comparison is the point of the exercise. Jiuzhang 4.0 produces a single output in 25.6 microseconds [20]. The researchers estimate that reproducing this with the best-known classical spoofing algorithm on the El Capitan supercomputer would take more than 10^{42} years.

That claim has been through review. The result was first posted as a preprint in August 2025 and published in Nature in May 2026 [20].

The progression is steep. The original Jiuzhang device in 2020 detected up to 76 photon clicks in a 100-mode circuit, published in Science, and was the first claimed photonic quantum computational advantage [22]. Jiuzhang 3.0 reached 255 photons [21]. Version 4.0 reached 3,050.

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